

Thiago Pari

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EDUCATION

Northeastern University

M.S., Robotics GPA: 3.7/4.0

Boston, MA

Expected May 2027

Coursework: Computer Vision, Robotic Science and Systems, Mobile Robotics, Robotics Sensing and Navigation, Machine Learning

New York University

B.A., Economics and Mathematics GPA: 3.6/4.0

New York, NY

May 2025

Coursework: Numerical Analysis, Linear Algebra, Differential Geometry, Fourier Analysis, PDEs, Mathematical Modeling

EXPERIENCE

Archenaux Robotics

Somerville, MA

Robotics Engineer Co-op, Underwater Perception & 3D Mapping

May 2026 – Sep 2026

- Owned 3D mapping architecture for a hull ROV; ruled out visual SLAM (median 51 ORB keypoints/frame), moving to sonar + IMU
- Traced sonar SLAM failure (0–6% of motion recovered) to vertical-mount bias, not noise; drove a redesign (0.2–0.3 m ATE in sim)
- Audited 8 pool rosbags and found 4 data defects, including an IMU frame bug (FRD vs. FLU) that reversed every lap reconstruction
- Wrote and deployed ROS 2 drivers for the Ping360 sonar and VN-100 IMU; found a serial bottleneck slowing sonar sweeps $\sim 20\times$
- Trained a real-time U-Net hull-inspection segmenter (0.50 mIoU, 247 FPS) after rejecting datasets that labeled red paint as rust
- Built Isaac Sim tools: a domain-randomized Replicator synthetic-data pipeline (biofouling) and a rosbag-calibrated sonar simulator

Citi

New York, NY

Summer Financial Analyst

Jun 2024 – Aug 2024

- Automated daily reporting with Excel VBA macros, cutting processing time by 50%

PROJECTS

Underwater SfM Degradation Study | Python, COLMAP | github.com/thiogopari/underwater-sfm-degradation Jul – Sep 2026

- Built a physics-based underwater renderer (measured Jerlov optics, sensor noise) and scored COLMAP poses over 64 seeded runs
- Showed SfM keeps one map to optical depth 2.4, splits at 2.7, and loses cameras from 3.2; color correction does not move it
- Benchmarked ALIKED + LightGlue vs. SIFT in turbid water: $5\times$ the points and 14 vs. 10–12 registered cameras, at $6\times$ the CPU time

Bayesian Skill Gate for VLA Robots, Nebius $\times$ NVIDIA Hackathon | Python, Isaac Sim, GR00T N1.7 Oct 2026 – Present

- Building a gate that certifies GR00T N1.7 VLA skills once Isaac Sim trials show $P(\text{success} \geq 0.8) \geq 0.95$; revokes on drift (team)

Multi-Sensor Fusion for Mobile Robot Localization | ROS 2, Python, EKF / UKF Nov 2025 – Jan 2026

- Built a 50 Hz ROS 2 EKF / UKF fusing odometry, IMU, and simulated GPS in Gazebo; cut RMS error 28.8% vs. dead reckoning
- Simulated 15 s GPS outages and wrote a trimmed, multi-resolution ICP scan-matching update for 2D LiDAR

Robustness Analysis of Convex Bundle Adjustment | C++, Python, SDP Nov 2025 – Dec 2025

- Stress-tested XM, a convex SDP bundle-adjustment pipeline (team); COLMAP feature matching, not the solver, was the bottleneck

12-DOF Quadruped Robot | Python, MuJoCo, Raspberry Pi, Fusion 360 Oct 2025 – Present

- Designed a 3D-printed 12-DOF quadruped (Raspberry Pi, IMU, I2C servos) that stands stably; developing IK and gaits in MuJoCo

SKILLS

Computer Vision & SLAM: OpenCV, COLMAP / SfM, GTSAM, KISS-ICP / GICP, Open3D point clouds, EKF / UKF sensor fusion

Deep Learning: PyTorch, semantic segmentation, object detection (YOLO), foundation models (DINOv2, AnyLoc), synthetic data

Robotics & Simulation: ROS 2 (rosbag2, DDS, TF, MAVROS), Isaac Sim / Replicator, Gazebo, MuJoCo; sonar, IMU, GPS, cameras

Programming & Tools: Python, C++, MATLAB, SQL, Git, Linux, Docker, pytest, GitHub Actions CI, Claude Code; Fusion 360

Languages & Interests: English, Spanish; Ironman training, chess (1600 rapid), guitar